Retail Inventory Analysis Report

Executive Summary

Analysis of 73,100 records (2022–2024) shows strong revenue performance but highlights significant inventory optimization opportunities. Data quality is excellent with no missing or duplicate values. Forecast accuracy is high (94.11%), indicating reliable demand planning, while overstock remains the primary operational issue.

Data Quality

73,100 records, 15 original variables (27 after feature engineering), zero missing values, and zero duplicates. This provides a reliable foundation for business decision-making.

Key Performance Indicators

Total Revenue: $550.23M
Revenue After Discount: $494.97M
Revenue Lost to Discounts: ~$55.26M
Units Sold: 9.98M
Orders: 8.04M
Average Selling Price: $55.14
Average Discount: 10.01%

Inventory Performance

Average Inventory: 274.47 units.
Average Inventory Turnover: 0.50, indicating relatively slow inventory movement.
Procurement Efficiency: 1.76 on average, with considerable variation across products.

Inventory Status

Low Stock: 3.54%
Overstock: 47.93%
Normal Stock: 48.53%

Business Interpretation: Nearly half of inventory is overstocked, tying up working capital and increasing storage costs.

Demand Forecast

Forecast accuracy reached 94.11%, indicating that demand forecasting is highly reliable. The operational challenge lies in purchasing decisions rather than forecasting quality.

Correlation Analysis

Revenue is strongly influenced by Units Sold (0.80) and Demand Forecast (0.80). Inventory Turnover (0.58) and Procurement Efficiency (0.51) also contribute positively. Inventory Difference (-0.23) negatively impacts revenue, suggesting excess inventory reduces financial performance.

Dependent and Independent Variables

Dependent Variable: Revenue.
Independent Variables: Units Sold, Demand Forecast, Inventory Turnover, Procurement Efficiency, Inventory Difference, Discount, Price, and Competitor Pricing.

Business Insights

Best Region: East
Best Category: Furniture
Best Season: Autumn
Best Weather: Sunny

Recommendations

1. Reduce overstock using forecast-driven purchasing.
2. Rely on the 94.11% forecasting model for replenishment decisions.
3. Review discount strategy to maximize profitability.
4. Prioritize Class A inventory and avoid stockouts.
5. Improve inventory turnover through optimized replenishment.
6. Allocate more inventory to the East region and before Autumn for Furniture products.
7. Immediately replenish the 93 critical items identified in the procurement analysis.

Conclusion

The company demonstrates strong sales performance supported by excellent forecasting capabilities. However, inventory optimization represents the greatest opportunity to reduce costs, improve cash flow, and increase profitability. Management should prioritize inventory optimization, procurement refinement, and strategic allocation of stock.